

Sudipta Mandal

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Working Experience

Software Engineer (L3), [Cargo Stream](#) (SaaS FreightTech, Vilnius, Lithuania, Remote), Nov 2025 - Present

- Developing a dynamic document-processing system that parses diverse logistics and invoice PDFs from multiple partners using Laravel, Text Extraction, Pattern Recognition, Data Analysis, and Context Engineering.
- Building a large number of flexible, layout-aware PDF parsers capable of handling inconsistent invoice structures across different companies with different languages.
- Extracting tabular and irregular data using custom logic, fallback rules, and pattern-matching techniques.
- Transforming raw PDF output into structured JSON and CSV formats for automated downstream workflows.
- Ensuring data accuracy by implementing normalization pipelines and format-agnostic extraction strategies.
- Working closely with the CTO to optimize data ingestion flows and improve document-processing performance.
- Enhancing the scalability and robustness of the parsing module to support increasing document volumes.

Senior Software Engineer, [Sports Fixtures](#) (Project-Based, Pattaya, Thailand, Remote), Oct 2025 - Jan 2026

- Built and maintained SportsFixtures.net from scratch using Node.js, Strapi v5, PostgreSQL, Neon DB, Redis, and Cloudinary; deployed and managed production infrastructure on Railway with centralized logging, monitoring, and environment management.
- Designed and optimized 20+ relational database models and advanced multi-model filtering/query systems, improving large-scale data retrieval performance by 40%.
- Implemented secure authentication and authorization features, including SMTP OTP verification, Google OAuth, and Facebook OAuth, improving login reliability and reducing authentication friction.
- Integrated complete SportsDB v1/v2 APIs and multiple third-party News APIs; developed automated cron job schedulers and background workers that reduced API synchronization downtime by 70%.
- Built real-time live sports score functionality using WebSocket connections and optimized backend performance with Redis and Node cache strategies, reducing redundant API/database load by 50%.
- Leveraged AI-assisted context engineering and agentic workflow automation to accelerate development workflows and improve engineering productivity by 50%.

Software Engineer (L1, L2), [Eucaps AB](#) (SaaS Fintech, Stockholm, Sweden, Remote), Dec 2021 - Oct 2025

- Built SaaS fintech features using Node.js, Express.js, Laravel, React/Next.js, Apollo/GraphQL, and REST APIs; leveraged AWS (Lambda, EC2, Amplify, S3), Docker, and Jest, with MySQL, PostgreSQL, and DynamoDB.
- Designed secure REST APIs (JWT + HMAC) enabling external partners to access selective database info; served 1K+ daily requests and cut integration time by 40%. Created clear and structured API documentation.
- Built backend pipelines to process 200MB+ [S&P](#) financial datasets (YoY, Margin, LTM) with cronjob schedulers, delivering low-latency GraphQL APIs that improved data accessibility for analysts and frontend functionalities.
- Integrated external APIs ([Pinpoint](#), [Inderes](#)) with cronjob schedulers, automating ingestion of 1K+ posts daily.
- Created scalable backend pipeline for 3K+ companies' financial charts & geo-maps with advanced filtering.
- Integrated [Stripe](#) payment and [Trulioo](#) KYC, enabling secure onboarding and seamless payment processing.
- Created backend pipelines to process data for 3K+ companies' events with services from [S&P](#) Capital IQ APIs.
- Leveraged GPT-3.5 + DALL·E to auto-generate hashtags and visuals for 3K+ company profiles, boosting analyst engagement by 25% and saving 5+ hrs/week in manual content preparation.

Junior Developer, [NSUPS](#), (Voluntary, Edtech, Dhaka, Bangladesh) Oct 2020 - Nov 2021

- Implemented an automatic crawler to fetch data from 10+ online judges to track the solve count frequency of more than 100 users. Created backend structure for a system managing the participants of NSUPS Bootcamp.
- Designed optimized database schemas and query structures to improve data consistency and backend maintainability.
- Collaborated with team members to develop scalable backend features and support internal university programming events.

Personal Projects

Text Analyzer [\[Web App\]](#) [\[Live Link\]](#)

- Developed an online platform to analyze text data with auto-generated reports. Implemented CRUD for text, with each browser acting as a unique user. Ensured fast, lightweight analysis. Adopted by **10+** active users.
- Built an interactive UI, optimized analysis, and deployed using Node, Express, TS, React, Vite, MUI, and TDD.

DSE SME Data Scraper [\[Script\]](#)

- Engineered an automated Node.js script tailored to reliably scrape daily SME company data from the Dhaka Stock Exchange (DSE).
- Implemented data transformation logic to parse raw HTML tables and seamlessly export the extracted financial records into structured CSV files for downstream analysis.

Product Pouch [\[Web App\]](#)

- Developed a full-stack interactive platform featuring complete product buy, sell, and rental functionalities.
- Built using Node, Apollo Express, Prisma ORM, and PostgreSQL to handle complex DB relations & user data.
- Designed a responsive and modern frontend utilizing React, Next.js, and Mantine UI, fully integrated with GraphQL queries and mutations.

Research & Thesis

Social Media Opinion Mining Based on Bangla Public Post of Facebook [\[Research\]](#)

- Focus: Sentiment analysis of Bangla text from Facebook public posts (Both Male & Female Celebrities)
- Coordinated data cleaning methods, data noise reduction, categorized, and annotated into 11 subcategories
- Created a dataset of 11K Bangla comments by using a **semi-automatic crawler** to fetch from Facebook
- Used state-of-the-art ML techniques (TF-IDF vectors, uni-bi, tri-gram, confusion matrix, etc) to analyze data
- Generated text clouds based on 3 main categories and 11 subcategories for the visual representation of data
- **Skills learned:** Python, Google Colab, traditional ML algorithms, LaTeX, etc.

Education

North South University (NSU), BSc in Computer Science & Engineering, **Graduated:** Fall, Aug 2021

- Cumulative GPA: 3.53/4.0 (88%+), CSE Major Cumulative GPA: 3.62/4.00 (90%+)
- Graduated with distinction: Cum Laude

Competitive Programming & Problem Solving

- **ACM ICPC Dhaka Regional (2019-2020):** Ranked [97th](#) among 190 teams (top 50%). Team: [NSU_0011](#)
- **National Collegiate Programming Contest (NCPC), MIST (2020-2021):** Ranked top 100 teams.
- **Intra NSU Programming Contest 2018:** Ranked [10th](#) among 46 individuals
- Organized, Judged, Setted problems for multiple contests organized by NSU & NSUPS. [Article](#)
- Solved **650+ problems** in online judges. **Codeforces rating:** [Specialist, 1426](#) (Max) **Handle:** Mr_Mandal

Leadership & Communication Skills

- **Coordinator** of Junior Programming Training Program, **Mentored** 50+ students. **Problem Setter & Judge:** Intra NSU Programming Contest [Spring 2023](#), NSUPS Bootcamp Graduation Contest [Season 07](#).
- Invigilate 3+ **ICPC** preliminary contests at North South University. PR officer in NSUPS'20 (Attendees: 1000+).
- **Hosted** an online [live session](#) on Facebook based on a programming community (500+ participants).
- English proficiency test result: IELTS Speaking **Band 7**

Skills

Backend: Laravel, Node.js, Express.js, Nest.js, REST, Django, GraphQL | **Frontend:** React, Next.js, Vite, MUI, CSS | **Databases:** MySQL, PostgreSQL, Dynamo, Mongo, Prisma | **Cloud:** AWS (Lambda, EC2, Amplify, S3), Docker, GCP | **Languages:** JavaScript, TypeScript, PHP, C/C++, Java, Python, SQL | **Tools:** Git, Jira, Confluence, Jest, TDD, CI/CD |